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Vertical farming economics: crop performance targets for cost-competitive vertical farming

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Vertical farming may offer a scalable solution to land, water, and climate constraints in agriculture, yet its viability hinges on reducing energy and capital costs. Despite these promises, the economic competitiveness of VF is strongly contested. High energy use, labour requirements and investment costs raise concerns about both economic feasibility and environmental sustainability. Vertical farming is a form of controlled-environment agriculture in which light, temperature, humidity, CO₂ concentration, irrigation, fertilisation and crop management can, in principle, be adjusted independently, enabling reproducible cultivation recipes that deliver consistent yield and quality. In this study, a crop-physiological source–sink model (the Plant Balance Model, PBM) is combined with an energy and techno-economic framework to analyse lettuce and tomato production in a modelled vertical farm. Three sink scenarios (Current, Current+, NextGen) are evaluated together with six levels of intercepted light-use efficiency ($LUE_{INT} = 0.8–1.8 \text{ g DW mol}^{-1} \text{ PAR}$), resulting in 18 scenarios per crop. Production costs are compared with Dutch greenhouse benchmarks from KWIN 2023. Results show that cost reduction depends primarily on increasing effective daily light integral and converting light efficiently into biomass through higher light interception and LUE_{INT} . As source capacity increases, production rapidly becomes sink-limited, making sink capacity and crop cycle time the next critical constraints. Under advanced but conceptually achievable performance targets, lettuce can approach Dutch greenhouse benchmark costs, whereas tomato requires both accelerated crop development and sustained high LUE_{INT} levels not yet realised at full-cycle scale. The framework defines crop performance targets rather than predicting realised commercial outcomes.

KEYWORDS

cost price, controlled environment agriculture, laminar airflow, lettuce, light use efficiency, techno-economic modelling, tomato, vertical farming

1 Introduction

Vertical farming (VF) can produce high-quality, residue-free crops while enabling year-round cultivation in controlled environments (SharathKumar et al., 2020). Closed-loop systems recycle water and nutrients, reducing leakage, while urban proximity shortens supply chains, lowers spoilage, and enhances food safety and shelf life (Van Delden et al., 2021). VF can also be resilient infrastructure, capable of buffering food systems against climatic extremes, pandemics, and supply-chain disruptions (Benke and Tomkins, 2017; Despommier, 2024).

However, despite these potential advantages, the commercial viability of VF has been disputed. High energy use, labour requirements and investment costs raise concerns about both economic feasibility and environmental sustainability (Banerjee and Adenauer, 2014; Stanghellini and Katzin, 2024). In particular, energy use has been identified as a major barrier and a contributing factor in several high-profile commercial failures within the sector (Pennisi et al., 2025).

From an engineering perspective, VF can be treated as a controllable production system in which cost reduction depends on identifying and optimising the dominant biological and technical constraints (Kaiser et al., 2024). Greenhouse horticulture demonstrates that increasing control over the growing environment can substantially increase productivity compared with open-field cultivation. By buffering crops from weather variability and enabling active control of temperature, CO₂ and water supply, greenhouse systems achieve higher and more stable yields per unit area (Heuvelink et al., 2025; Stanghellini and Katzin, 2024). VF extends this principle by placing cultivation in fully enclosed environments with artificial lighting, thereby minimising external disturbances and allowing environmental variables to be controlled with high precision to control and improve crop performance (Benke and Tomkins, 2017; Despommier, 2024).

In this study, we explore the effect of improvements in crop performance on production costs in vertical farms. To address this, a crop physiological model, the Plant Balance Model (PBM; Meeuws et al., 2025), is integrated with a techno-economic framework. The PBM represents yield formation as the balance between assimilate supply (“source”) and sink capacity. In controlled environments, increasing light supply and potential photosynthesis can rapidly shift crops into sink limitation, making sink capacity, crop architecture and crop cycle time central system constraints. This source–sink framing is essential for identifying which controllable variables in a vertical farm offer the highest leverage for reducing cost per kilogram.

In greenhouse and field systems, yields are typically source-limited. In contrast, the controllable light, CO₂ availability, and climate in VF can shift crops toward sink limitation. Under such conditions, increasing sink capacity increases yield only if assimilate supply (source) is adjusted accordingly. Otherwise, additional sink strength does not translate into higher output (Meeuws et al., 2025).

We apply this framework to lettuce and tomato, exploring eighteen production scenarios per crop. These eighteen scenarios comprise three configurations—Current, Current+, and NextGen—with six intercepted light use efficiencies ($LUE_{INT} = 0.8, 1.0, 1.2, 1.4, 1.6, \text{ and } 1.8 \text{ g dry weight (DW) mol}^{-1} \text{ photosynthetically active radiation (PAR)}$) (see [Supplementary Table S3](#) for a full overview of scenario settings). The cultivation strategies represent distinct structural and biological constraint regimes rather than a sequential development pathway. LUE_{INT} is therefore varied independently across all scenarios to isolate interactions between physiological efficiency and system constraints.

The combination of three sink scenarios and six LUE_{INT} levels results in 18 distinct production scenarios. These scenarios are calculated for both lettuce and tomato, enabling systematic comparison of how improvements in physiological efficiency and sink capacity affect yield formation and cost structure in VF systems. Yield is determined by the most limiting of source or sink capacity, and lighting (PPFD and resulting DLI) is adjusted accordingly. These scenarios should be

interpreted as conceptually independent design cases rather than as sequential development stages or a projected time path.

- ‘Current’ scenarios represent a state-of-the-art vertical farm assembled from individually demonstrated components and best-practice operating assumptions.
- ‘Current+’ scenarios depict a configuration characterised by an increased sink capacity using relatively direct crop-structural adjustments: higher planting densities for lettuce and tomato, increased fruit number per truss for tomato, and improved canopy light interception.
- ‘NextGen’ scenarios depict an independent design case characterised by a combination of crop-structural and crop-physiological assumptions. Crop development is accelerated by increasing cultivation temperature to 28 °C, consistent with established temperature–development relationships (Bensink, 1971; de Koning, 1994; Yan and Hunt, 1999), with PBM relations used to determine the resulting changes in sink dynamics (Meeuws et al., 2025). This concept represents a design target that implies new cultivars capable of sustaining accelerated development at elevated temperature without physiological disorders or quality penalties.

Costs are benchmarked against Dutch greenhouse production using KWIN 2023 (the Quantitative Information on Dutch Greenhouse Horticulture) (Raaphorst and van Tuyll, 2023). KWIN provides transparent, empirically grounded cost accounting and regularly updated technical assumptions, making it a robust reference for method alignment and interpretation of cost drivers. The greenhouse benchmark is not used to frame competition between vertical farms and greenhouses, but to provide a rigorous accounting reference. For the vertical farm cost model, assumptions and accounting approaches were aligned with KWIN practice as closely as possible. Both the greenhouse reference and the vertical farm design are evaluated at a cultivation area of 40,000 m² to enable consistent scaling of investment and operating costs.

2 Methods

Here, a techno-economic framework is developed in alignment with the KWIN 2023 (Raaphorst and van Tuyll, 2023), the standard Dutch horticultural reference providing regularly updated technical and economic benchmarks for greenhouse crops. The analysis reflects an average year of operation at steady state, situating the modelled vertical farms in the context of the Netherlands and enabling direct benchmarking against Dutch greenhouse production.

Physiological crop performance was calculated using the Plant Balance Model (PBM; Meeuws et al., 2025), which estimates attainable yield as the minimum of source- and sink capacity. Source-limited yield is determined by intercepted photosynthetically active radiation and light-use efficiency, while sink-limited yield is determined by planting and fruit density and temperature-dependent crop growth cycles. Full model equations and parameterisation are provided in the [Supplementary Methods](#). In the present study, the PBM is applied as a mechanistic framework for analysing system constraints and design targets, rather than as a validated predictive model of near-term commercial performance.

Intercepted light-use efficiency (LUE_{INT}) is defined as grams of dry matter produced per mole of intercepted photosynthetically active radiation ($g\ DW\ mol^{-1}\ PAR$). In the PBM, canopy light interception, physiological conversion efficiency (LUE_{INT}), and harvest index are parameterised independently, allowing physiological efficiency to be separated from canopy structure and biomass partitioning (Meeuws et al., 2025).

Much of the literature reports light-use efficiency relative to incident radiation (LUE_{INC}), which conflates interception and conversion efficiency [e.g., Jin et al. (2023)]. Reported LUE_{INC} values must therefore be converted to LUE_{INT} before being used as model inputs. Reported efficiencies span wide ranges because interception is often below 100% and because studies differ in dry-matter accounting (Carotti et al., 2021; Jin et al., 2023; Joshi et al., 2017; Pennisi et al., 2019) (see Supplementary Methods ‘Light-Use Efficiency’).

This study explores LUE_{INT} values from 0.8 to 1.8 $g\ DW\ mol^{-1}\ PAR$ are explored (Table 1). This range does not reflect average operational performance but represents observed, best-practice and speculative targets. A lower bound of 0.8 $g\ DW\ mol^{-1}$ is adopted because these values have been experimentally achieved for both lettuce (Carotti et al., 2021; Jin et al., 2023; Joshi et al., 2017; Pennisi et al., 2019) and tomato (Karpe et al., 2024; Saito et al., 2020) for whole-cycle crop efficiency. Values above approximately 1.4 $g\ DW\ mol^{-1}\ PAR$ are not currently realised as full-cycle averages in commercial vertical farms and are included to define physiological design targets rather than to imply present-day feasibility.

Lettuce and tomato are assessed as model crops as they represent leafy and fruiting production systems and both are among the most commonly studied VF crops (e.g., Jin et al., 2023; Kaiser et al., 2024). Lettuce was benchmarked against ‘Lettuce tricolor, hydroponic with lighting reference’ greenhouse data from KWIN 2023 (Raaphorst and van Tuyl, 2023), and tomato against ‘Tomato, cocktail vine w40’. The modelled vertical farm was defined with a net growing area of 40,000 m^2 , equal to the size of the greenhouse KWIN 2023 reference (Raaphorst and van Tuyl, 2023), enabling comparison between the systems.

The vertical farm was assumed to be housed in a building with a total height of 25 m. Layer numbers were derived from assumed layer heights of 1.25 m for lettuce and 2.5 m for tomato, resulting in 20 and 10 stacked layers, respectively. These layer heights include the crop

zone, lighting, and service space. Furthermore, this study assumes climate uniformity, which can be achieved by implementing horizontal laminar airflow system with air handling units and a heat pump (Agati et al., 2024). These systems can further allow the use of proven mobile gully systems for lettuce and gully systems for tomato and reducing capital requirements through reliance on commercially available horticultural technology.

To accommodate the airflow system, associated plenums, access paths, and other operational infrastructure, we allocate 15% of the gross cultivation footprint to non-growing space, resulting in a net-to-gross growing-area ratio of 85%. This ratio is lower than that of the greenhouse benchmark, where 96% of the footprint is allocated to cultivation area and 4% to access paths and operational space (Raaphorst and van Tuyl, 2023). When production is expressed per unit gross growing-area footprint, the net cultivation area is converted to the corresponding gross growing-area footprint using the net-to-gross ratio and adjusted for the number of cultivation layers. Additional handling areas were excluded from this metric and were only considered in the capital expenditure calculations.

2.1 Energy modelling

Energy demand was calculated using an integrated heat and mass balance approach that accounts for lighting, heating, ventilation, air conditioning (HVAC), and dehumidification. The heat pump coefficient of performance (COP), fan- and LED efficiency, and spectral efficacies were included explicitly.

The grow space was assumed to be adiabatic (i.e., without thermal leakage across the building envelope), reflecting the highly insulated construction typical of vertical farms. This assumption is consistent with previous modelling studies that represent the growing environment as a closed system with no energy exchange with the exterior climate (Graamans et al., 2018). Consequently, heat transfer with the external environment was not considered in the energy balance. Furthermore, the model resolves two operational states, the photoperiod and the dark period, which together form a 24 h cycle. For each scenario, based on its specific PBM input values, the energy balance was calculated separately for the photoperiod and dark period and then extrapolated over all growing days to determine annual electricity use. A steady-state was assumed for each 24 h cycle throughout the year.

All flows in the energy balance are expressed as instantaneous power (kW). Electrical loads from lighting and fans can be directly converted to electricity use (kWh) by multiplying by their operating hours in each period. Thermal loads from heating, cooling, and dehumidification are met by a heat pump with a fixed COP of 4 for heat production, and a cooling COP of 3. These values are representative of modern commercial heat-pump systems, for which COP values around 3–5 are commonly reported. The corresponding electricity demand is calculated by dividing the thermal heating and cooling loads by their respective COP values.

2.1.1 Energy balance of an indoor farm during the lighting period

The energy balance model is based on the patented design of Meeuws et al. (2019), which describes the airflow cycle through the growing system. The cycle is similar for both the photoperiod and the dark period, but differs in setpoints for evaporation, vapor pressure

TABLE 1 Overview of the included LUE_{INT} values.

LUE_{INT} ($g\ DW\ mol^{-1}\ PAR$)	Category	Description
0.8–1.0	Observed (whole cycle)	Experimentally achieved for lettuce and tomato
1.2–1.4	Best-practice (optimised)	Reported under optimised conditions, but not consistently demonstrated as full-cycle commercial averages
1.6–1.8	Speculative design target	Not realised at full-cycle commercial scale; included to define physiological targets, not present-day feasibility

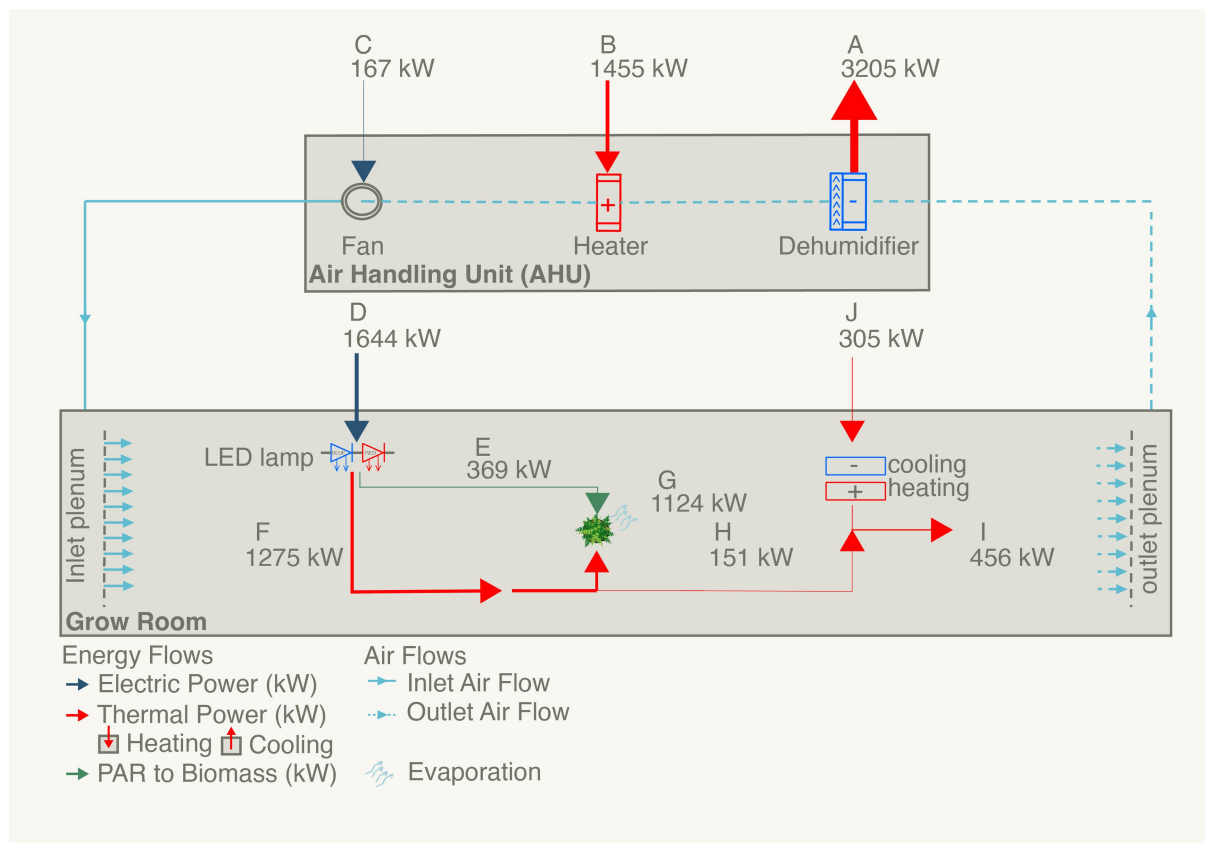


FIGURE 1

Schematic of energy flows (A–J) of scenario Current ($LUE_{INT} = 1.2$) lettuce during the photoperiod (see the [Supplementary material](#) for all scenarios for both tomato and lettuce and both during photo- and dark period). Values are given in kW. External demands are A (cooling), B (heating), and J (additional heating/cooling), which are thermal loads met via a heat pump (electric demand = thermal load for heat production/COP and electric demand = thermal load for cold production/cooling COP), and C (fan) and D (lighting), which are direct electric loads. E–I represent internal or crop-driven transfers, including photosynthetic energy storage in biomass (E), latent heat for transpiration (G), and intermediate thermal exchanges (F, H, I).

deficit (VPD), and lighting (see [Figure 1](#) for photoperiod energy flows in the Current ($LUE_{INT} = 1.2$) scenario; for other scenarios and dark period flows see [Supplementary Figures S3–S50](#)). The main distinction is that during the dark period the LED system is off, and no lighting energy enters the balance. In both cases, the air handling unit (AHU) supplies air at the desired inlet temperature [$\theta_{(in)}$] and vapor pressure deficit [$VPD_{(in)}$], with absolute humidity [$x_{(in)}$] that matches these conditions.

The conditioned air enters the grow room through the inlet plenum. Inside, it passes the LED system, which draws power (D) from the grid and converts it into photosynthetically active radiation (PAR) (E) during the photoperiod (see [Figure 1](#) for an example of corresponding flows). A portion of this absorbed PAR is used by the plants to produce dry matter. Combined with water and minerals, this results in fresh plant biomass. The remaining power ($F = D - E$) is released as heat. Some of this (G) is used for plant transpiration, increasing the air's absolute humidity [$x_{(out)}$] as it leaves the grow room. The residual heat ($H = F - G$) can be positive or negative. If H is negative, heat from the surrounding air is used to sustain transpiration, thus lowering air temperature. If H is positive, surplus heat raises the air temperature. The air leaves the grow room at the same $VPD_{(in)}$ it entered with, but at a higher temperature and humidity. Additional heating or cooling power (J) may be added or removed to achieve the desired outlet temperature [$\theta_{(out)}$]. The return air flows back to the AHU which first passes through the cooling coil, where heat (A) is

removed and moisture condenses until humidity matches $x_{(in)}$ for the required supply temperature $\theta_{(in)}$ and $VPD_{(in)}$. The air then passes through the heating coil, where heat (B) is added to warm the air to $\theta_{(in)} - \Delta T$. Finally, it passes the fan, which adds heat (C) due to inefficiencies, raising the temperature by ΔT so that $\theta_{(in)}$ is restored, this completes the cycle. To have an equal average temperature in the grow room, the direction of the air is changed at regular intervals.

2.2 Cost modelling

Operational expenditures (OPEX) are comprised of electricity, labour, fertiliser, CO₂, seedlings, and growing media. Expenditures were set at national costs with an electricity price of €0.111 kWh⁻¹, labour of €22.50 h⁻¹, and CO₂ of €0.12 kg⁻¹ equal to the KWIN ([Raaphorst and van Tuyl, 2023](#)). Crop-specific costs for seedlings and growing media were also taken from the KWIN ([Raaphorst and van Tuyl, 2023](#)): €0.08 plant⁻¹ and €0.00578 plant⁻¹ for lettuce, and €3.00 plant⁻¹ and €0.86 m⁻² for tomato, respectively. All non-electricity OPEX values were scaled to the scenario-specific production outputs.

Capital expenditures (CAPEX) were estimated using KWIN ([Raaphorst and van Tuyl, 2023](#)) data where equivalent technology could be applied, covering items such as heating systems, CO₂ systems, irrigation, water storage, disinfection, gutters, transport systems, and computing equipment. VF-specific components, including the commercial building for the growing area, vacuum cooler (for lettuce),

air handling units (AHU), plenums, and multilayer facilities, were assigned assumed values informed by market research. Scenario-specific variation was incorporated for lighting, AHU, and heat pump capacity. Miscellaneous costs were set high, at 10% of total CAPEX (see [Supplementary material](#) for all expenditures).

For each CAPEX category, KWIn (Raaphorst and van Tuyl, 2023) provides low and high reference prices, along with depreciation and maintenance rates; these were scaled to the modelled VF configuration (see [Supplementary Table S4](#) for equipment specific maintenance and depreciation rates and related scale factor). The average of the low and high price estimates was used, multiplied by the category-specific depreciation and maintenance rates to determine annual costs. Interest was calculated as annual depreciation of $3.6\% \times 0.55$, following the KWIn's (Raaphorst and van Tuyl, 2023) method to account for interest only on the average book value. This approach follows KWIn's steady-state methodology, in which depreciation, maintenance, and interest are expressed as fixed percentages of CAPEX and reported as yearly costs, representing an average year of operation.

The cost price (€ kg^{-1} FW) was obtained by dividing the total annual expenditures (OPEX + CAPEX-derived costs) by the annual yield. Full parameter and input variable values are provided in [Supplementary Tables S1–S4A](#).

Although this study is parameterised for Dutch conditions to enable KWIn benchmarking, the modelling framework can be adapted to other crops and regions by replacing crop-specific PBM parameters and local cost modules. However, the quantitative outcomes reported here should be interpreted within the Dutch reference context.

2.3 Sensitivity- and uncertainty analysis

We combined a local one-at-a-time (OAT) sensitivity screen with a Monte Carlo (MC) uncertainty analysis. In OAT, each input was swept across its scenario range with others held at baseline; effects are reported as normalised sensitivities (see [Supplementary Tables](#)). In MC, inputs were sampled from predefined distributions (triangular

for bounded/elicited quantities, lognormal for strictly positive prices/intensities, fixed where appropriate) for 10,000 iterations per scenario. Input distributions were derived from three sources: literature-reported ranges, market and statistical data, and assumption-based uncertainty ranges where published information was unavailable. Assumption-based ranges were used to represent plausible variation around baseline values for engineering and operational parameters lacking empirical uncertainty estimates and were informed by commercial horticultural engineering practice in the Dutch context. Cost-price outcomes (€ kg^{-1} FW) are summarised as median [IQR] and 95% uncertainty intervals (UIs); full parameterisation of distributions and sources is provided in [Supplementary methods](#), [Supplementary Tables S5–S9](#).

3 Results

Modelled yields per m^2 of net growing area varied across the three configurations and increased with higher intercepted light-use efficiency (LUE_{INT}), from 78 to $330 \text{ kg FW m}^{-2} \text{ yr}^{-1}$ for lettuce (Current ($\text{LUE}_{\text{INT}} = 0.8$) to NextGen ($\text{LUE}_{\text{INT}} = 1.8$)) and from 49 to $369 \text{ kg FW m}^{-2} \text{ yr}^{-1}$ for tomato, compared with Dutch greenhouse benchmarks of 68 and $59 \text{ kg FW m}^{-2} \text{ yr}^{-1}$, respectively ([Figure 2](#)). When expressed per unit gross growing area footprint, accounting for multilayer stacking (20 layers for lettuce and 10 for tomato) and the net-to-gross conversion, production reached $5,617 \text{ kg FW m}^{-2} \text{ yr}^{-1}$ for lettuce and $3,133 \text{ kg FW m}^{-2} \text{ yr}^{-1}$ for tomato. These values reflect the land-use advantage of multilayer cultivation, which allows substantially more growing area to be accommodated within the same footprint.

Electricity intensity decreased with higher LUE_{INT} and the associated increases in yield. Lettuce fell from 7.83 to $2.31 \text{ kWh kg}^{-1} \text{ FW}$, and tomato from 20.64 to $6.69 \text{ kWh kg}^{-1} \text{ FW}$ across the evaluated scenarios (see [Figure 3](#)). Lighting remained the dominant energy consumer across all scenarios. However, as LUE_{INT} increased, its relative contribution diminished due to better biomass conversion. Climate

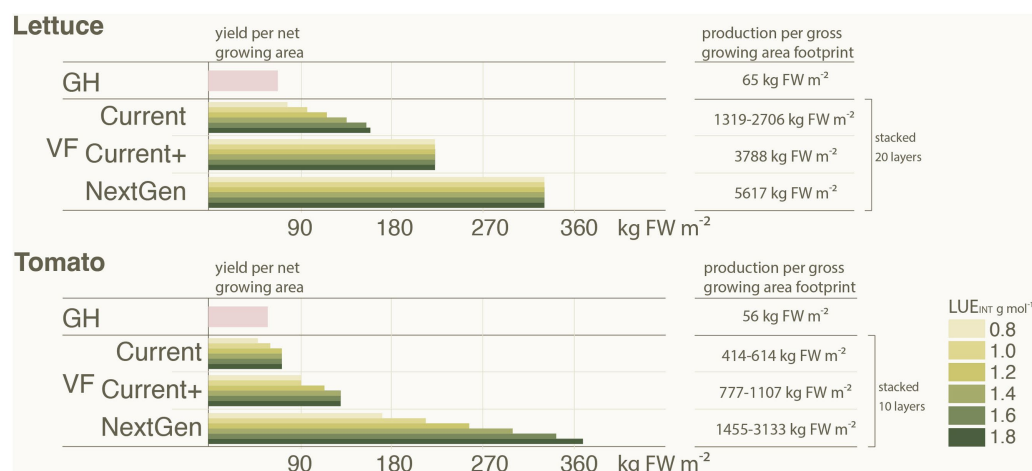


FIGURE 2

Annual yield per unit net growing area ($\text{kg FW m}^{-2} \text{ yr}^{-1}$) for lettuce and tomato under vertical farming (VF) scenarios and the greenhouse (GH) benchmark. Values shown in the right-hand panel indicate the corresponding production per unit gross growing-area footprint, calculated by accounting for multilayer stacking (20 layers for lettuce; 10 layers for tomato) and the net-to-gross growing-area ratio (85% for VF; 96% for GH). Additional handling area was excluded from this metric for both VF scenarios and the greenhouse benchmark.

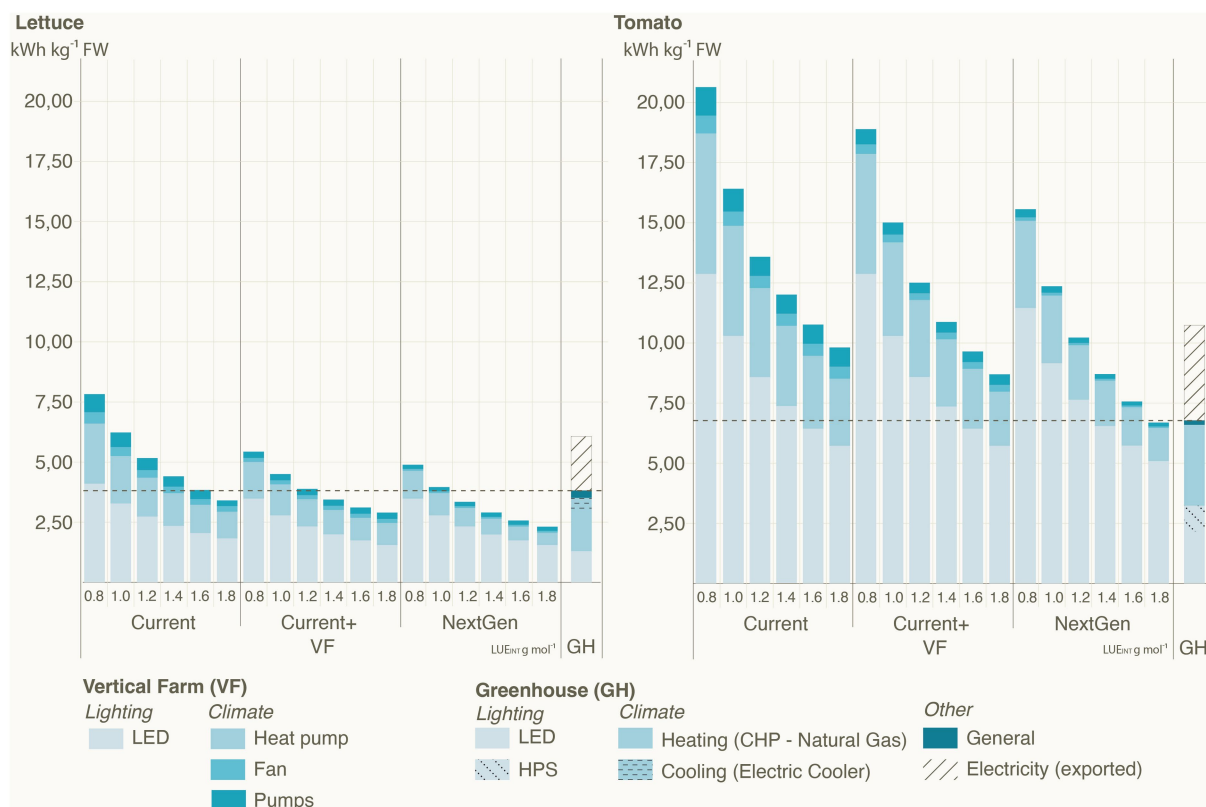


FIGURE 3

Overview of energy use per output (kWh kg⁻¹ FW) for both lettuce and tomato across the VF scenarios and the KWIN Greenhouse benchmark. Dashed lines represent the net greenhouse energy use, accounting for electricity exported to the grid (Raaphorst and van Tuyl, 2023).

control energy (for cooling, heating, and humidity regulation via heat pumps) also fell per unit output.

Relative to Dutch greenhouse references (3.81 kWh kg⁻¹ FW for lettuce; 6.78 kWh kg⁻¹ FW for tomato; not accounting for exported electricity), the baseline Current (LUE_{INT} = 0.8, 1.0, and 1.2) scenarios remain above the benchmark. For lettuce, increases in LUE_{INT} (≥1.4) reduced electricity intensity below the greenhouse benchmark across most configurations. For tomato, parity was achieved only under the NextGen assumptions at LUE_{INT} = 1.8.

Annualised capital cost per kilogram (depreciation, maintenance and interest) decreased as output increased across the evaluated scenarios (Figure 4): lettuce prices drop from €1.34 to €0.36 kg⁻¹ FW and tomato from €2.96 to €0.79 kg⁻¹ FW from Current (LUE_{INT} = 0.8) to NextGen (LUE_{INT} = 1.8). Lighting was the largest component in all scenarios, followed by climate systems (air handling units, heat pumps, plenum) and growing systems (e.g., gutters, multilayer structures); other infrastructure—including buildings, water and CO₂ delivery systems—contributed less. While some investment items shift between scenarios, most capital requirements change little, and their contribution per kilogram mainly declines through dilution via higher yields.

Relative to Dutch greenhouse references (€0.92 kg⁻¹ FW for lettuce; €0.57 kg⁻¹ FW for tomato), vertical farms were below the benchmark for lettuce in all scenarios except Current (LUE_{INT} ≤ 1.2), whereas the best tomato configuration remained ~37% higher (€0.78 vs. €0.57).

Total cost price varied substantially across the evaluated scenarios, from €3.26 to €1.67 kg⁻¹ FW (95% UI 2.85–4.08 to

1.49–2.02) for lettuce and from €5.84 to €2.04 kg⁻¹ FW (95% UI 4.77–7.70 to 1.67–2.69) for tomato (Figures 5A,B). Reductions primarily reflected dilution of fixed equipment costs over higher output, with additional decreases from lower electricity use per kilogram at higher LUE_{INT}.

For lettuce, all configurations except Current (LUE_{INT} ≤ 1.2) were below the reference of €2.35 kg⁻¹ FW. For tomato, even the highest-yielding scenario (NextGen, LUE_{INT} = 1.8) remained approximately 12% above the greenhouse reference (€1.82 kg⁻¹ FW). Further details on uncertainty are provided in Supplementary Tables S6–S7, Supplementary Figures S4–S5.

3.1 Sensitivity analysis

For lettuce (Current, LUE_{INT} = 1.2; Figure 6A), biophysical drivers dominate: increases in light interception, light-use efficiency, and photoperiod substantially reduce production cost, whereas higher investment cost and seedling price increase it; other inputs exert comparatively smaller effects. For tomato (Current, LUE_{INT} = 1.2; Figure 6B), cost sensitivity is primarily driven by light capture and conversion efficiencies and financing costs, followed by electricity price and lighting system efficiency, while crop structure (fruit number per truss and stem density) and labour cost have a more limited influence. Full elasticity values and scenario means are reported in the Supplementary Tables S8–S9. Monte Carlo uncertainty analysis [median (IQR), 95% intervals] confirms these hierarchies (Supplementary Tables S2–S3, Supplementary Figures S2–S3).

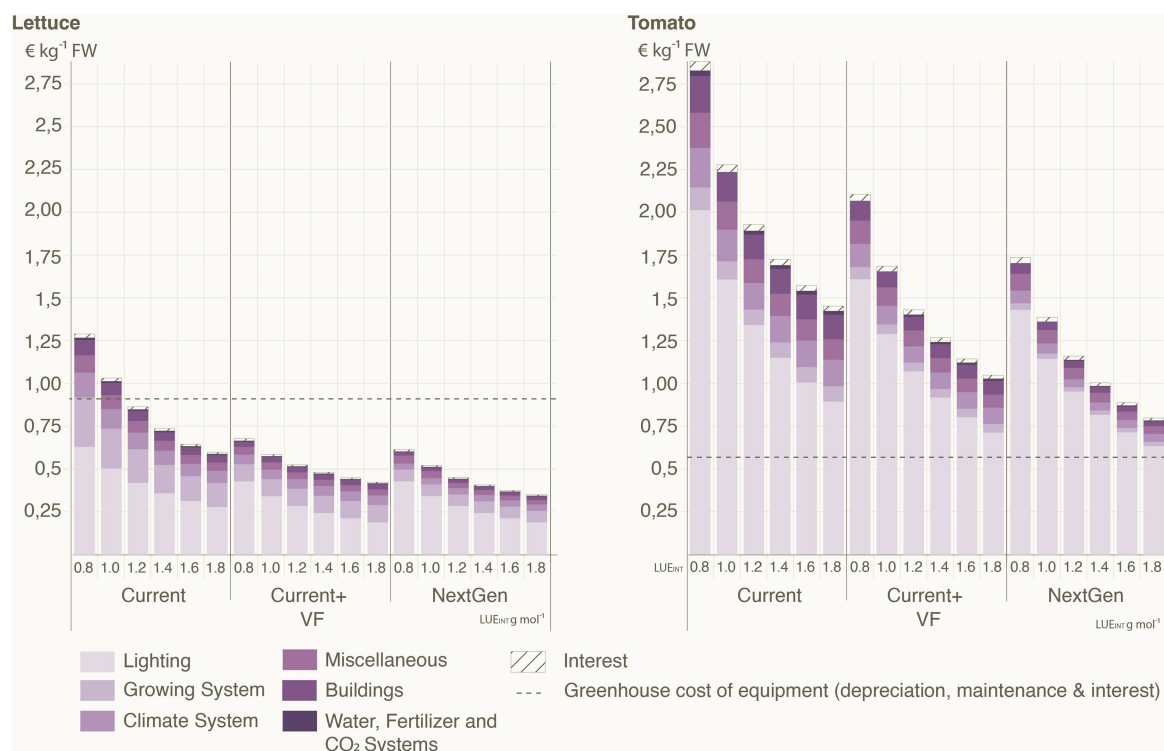


FIGURE 4

Overview capital cost per kilo of produce (€ kg⁻¹ FW), reflected as buildings plus equipment costs for depreciation, maintenance and interest, for lettuce and tomato, disaggregated by system component. Dashed lines indicate KWIN Greenhouse benchmarks (Raaphorst and van Tuyl, 2023).

4 Discussion

Applied system boundaries do not include other factors that provide advantages of VF, such as local-to-local supply that reduces refrigerated transport and downstream waste as VF produce has a longer shelf life and lower spoilage (Banerjee and Adenauer, 2014; Benke and Tomkins, 2017). We also do not include the prices of environmental externalities nor potential premiums for the produce. For example, pesticides are not used in VF settings—but may be used in greenhouse benchmarks—and therefore can command an organic premium. Ultimately, such sustainability attributes are also central to societal acceptance, as consumer studies show that willingness to adopt VF products depends strongly on perceived sustainability (Jürkenbeck et al., 2019; Perambalam et al., 2021). Improving the sustainability profile, particularly by lowering energy intensity and material input, remains key for both economic viability and societal acceptance.

Furthermore, land-constrained urban areas represent a particularly strategic opportunity. Van Delden et al. (2021) identify this niche as a key application domain for VF, an insight supported by the gross growing-area footprint production values reported in Figure 2. Across all scenarios, vertical farms achieved around 20–80 × higher production per gross growing-area footprint for lettuce and 7–50 × for tomato relative to Dutch greenhouses, primarily due to multilayer stacking and intensified production cycles. This underscores the strategic potential of VFs in high-density urban areas. For a fair land-use comparison that includes off-site land for energy supply, whenever VF energy use exceeds greenhouse energy use, the VF's attributed energy-production land should be scaled by the energy-use ratio (VF/GH)—× 1.36 for lettuce (current) and × 2.01 for tomato (current). Conversely, in scenarios where VF uses less energy per kg FW than

greenhouses, this factor is <1, further improving VF's land advantage (e.g., lettuce NextGen ($LUE_{INT} = 1.8$) × 0.61 ≈ 39% less energy land).

VF is frequently portrayed as excessively energy intensive—three- to sevenfold higher than advanced greenhouses (Blom et al., 2022; Graamans et al., 2018). Much of this apparent gap reflects assumptions used in earlier studies about LUE and LED system efficacy. Back-calculation of LUE_{INT} from those datasets indicates markedly lower canopy performance than contemporary systems. Using updated biological parameters and lighting performance, modelling yields substantially lower energy intensities while confirming that electricity remains the dominant operating cost (Figure 3).

When expressed on a fresh-weight (FW) basis, the energy intensities reported by prior studies exceed ours even before accounting for biological and technological updates. Graamans et al. (2018) report 247 kWh kg⁻¹ on a dry-weight (DW) basis. Using their 7% dry-matter (DM) assumption, this converts to 17.29 kWh kg⁻¹ FW, or more than twice higher than the lowest yielding lettuce scenario in this study (7.83 kWh kg⁻¹ FW). Under the 3.5% DM used here (Carotti et al., 2021; Jin et al., 2023), the same figure is 8.65 kWh kg⁻¹ FW, around 10% above our estimate.

Two assumptions dominate the remaining difference. First, light-use efficiency (LUE_{INT}), defined here as dry-matter produced per mol of intercepted PAR (g DW mol⁻¹ PAR; excludes harvest index), is lower in earlier work: values implied by the PPFD/photoperiod and reported productivity in Graamans et al. (2018) yield ≈ 0.51 g DW mol⁻¹ PAR. However (Jin et al., 2023) show that a LUE_{INT} of 1.20 g mol⁻¹ PAR can be achieved in modern VFs, ~58% higher than the value used in Graamans et al. (2018). Second, system-level LED electrical-to-photon efficiency (including drivers and optics) is assumed at ~53% in Graamans et al. (2018), whereas

A) Lettuce

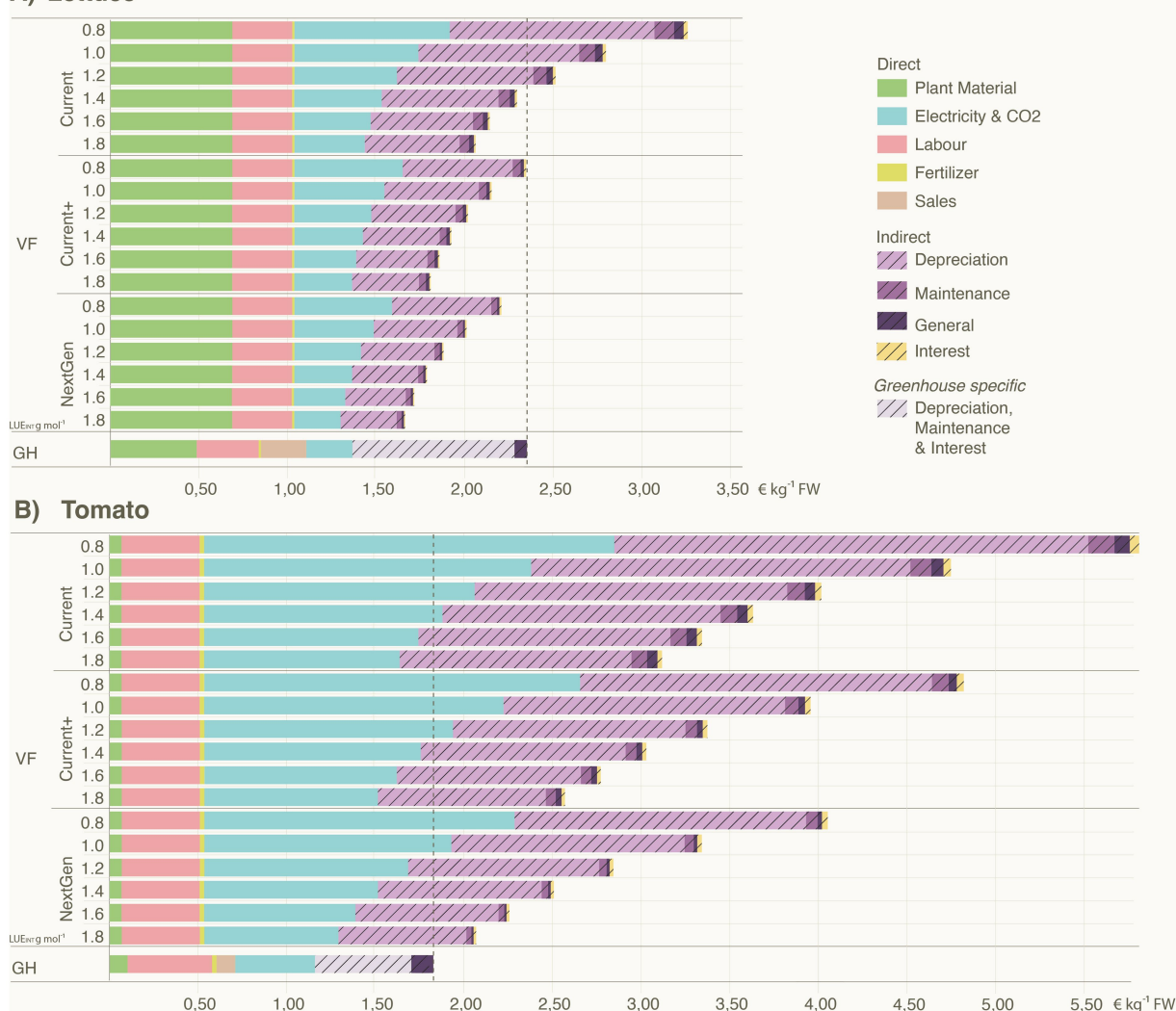


FIGURE 5

Total cost price per kg FW of produce for lettuce (A) and tomato (B) across all scenarios and KWIN Greenhouse benchmarks (Raaphorst and van Tuyl, 2023). Including disaggregated cost contributions for direct (plant material, electricity & CO₂, labour, fertiliser and sales) and indirect (depreciation, maintenance and other indirect costs) categories. For comparison to greenhouse cost price levels throughout the year (per 4-week period) see Supplementary Figure S1.

contemporary integrated systems typically achieve ~70%, and we assume 73% (Pattison et al., 2018; Supplementary material). Imposing $LUE_{INT} = 0.51 \text{ g DW mol}^{-1} \text{ PAR}$ and $LED = 53\%$ in our model raises the requirement to $15.61 \text{ kWh kg}^{-1} \text{ FW}$, 202% above our baseline and ~80% higher than the value derived from Graamans et al. (2018) after 3.5% DM alignment ($8.65 \text{ kWh kg}^{-1} \text{ FW}$). This underscores how conservative plant-level performance and outdated lighting efficacies inflate kWh kg^{-1} .

For Blom et al. (2022), the reported $14.8 \text{ kWh kg}^{-1} \text{ FW}$ likewise reflects low canopy performance. Back-calculating from their 49.2 W m^{-2} lighting input (no CO₂ supply due to a leaky container) implies a photon efficacy $\approx 3.0 \mu\text{mol J}^{-1}$, a DLI $\approx 12.7 \text{ mol m}^{-2} \text{ d}^{-1}$, and, using 3.5% DM and 95% HI, an $LUE_{INT} \approx 0.38 \text{ g DW mol}^{-1} \text{ PAR}$. This is ~55% below our baseline LUE_{INT} of 0.8, and consistent with our ~50% lower energy intensity (7.83 vs. $14.8 \text{ kWh kg}^{-1} \text{ FW}$) (Figure 3).

In our model, lower LUE_{INT} VF configurations are substantially more energy-intensive than Dutch greenhouses benchmark values, with the least efficient lettuce and tomato cases at approximately

two- and three times higher electricity use per kilogram, respectively. As LUE_{INT} and overall system optimisation increase, energy intensity declines sharply, such that highly efficient lettuce configurations outperform the greenhouse benchmark, while tomato converges towards parity only under the most optimised conditions. A key factor today is that Dutch greenhouses benefit from combined heat and power (CHP) installations that allow electricity export to the grid and thereby improve overall energy balances (Figure 3). However, the persistence of this advantage is uncertain. Government plans regarding energy taxation and CO₂ pricing have been delayed, with more clarity expected in the coming period (Ministry of Agriculture, Fisheries, Food Security and Nature, 2025). Under alternative energy pricing or carbon accounting regimes, the relative position of greenhouse horticulture may deteriorate. Moreover, the Dutch greenhouse sector represents a global best case in terms of infrastructure, climate suitability, and optimisation; these advantages are not universally available, and results are not directly generalisable to regions with less advanced greenhouse industries or less favourable production conditions.

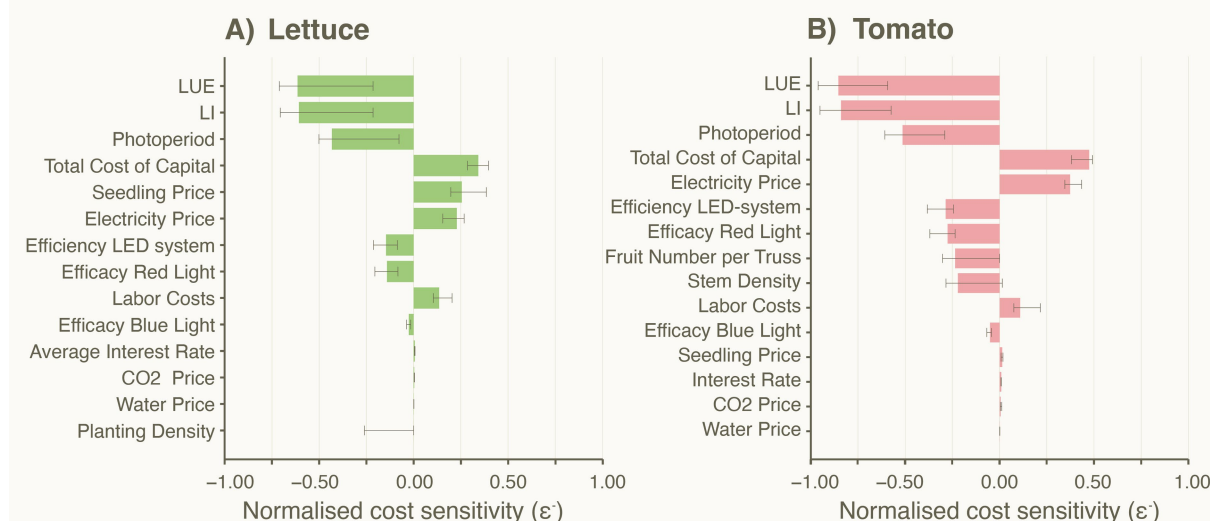


FIGURE 6

Local cost sensitivities in vertical farming. (A) Lettuce, Current scenario ($LUE_{INT} = 1.2 \text{ g mol}^{-1}$). (B) Tomato, Current scenario ($LUE_{INT} = 1.2 \text{ g mol}^{-1}$). Bars indicate elasticities for the Current scenario ($LUE_{INT} = 1.2 \text{ g mol}^{-1}$), while whiskers show the corresponding range across all scenarios. Detailed values are provided in the [Supplementary Tables S8–S9](#).

One area of interest in the literature for reducing VF costs has been dynamic lighting, the modulating of intensity or spectrum in order to lower overall energy cost (e.g., [Kaiser et al. \(2024\)](#)). However, our OAT elasticities indicate that, while such strategies can trim marginal energy spend, their standalone effect on cost price is modest relative to biological levers. A proportional 10% drop in electricity price is expected to reduce cost by 2.3% (lettuce) and 3.7% (tomato); the same 10% improvement in LUE_{INT} lowers costs by 5.9 and 7.5%, and a 10% increase in light interception (LI) yields 5.8 and 8.4% reductions, respectively (see Section 3.1). Thus, pricing-responsive lighting should complement, not substitute, higher-leverage improvements in LUE_{INT} , light interception (LI), and crop genetics/densities. Moreover, although our model caps LUE_{INT} at $1.8 \text{ g DW mol}^{-1} \text{ PAR}$, theoretical potentials $>2.1 \text{ g DW mol}^{-1} \text{ PAR}$ have been proposed ([Meeuws et al., 2025](#)), indicating scope for additional cost reductions.

Beyond dynamic lighting, improving diode efficacy is another frequent proposal. Our sensitivity elasticities indicate these levers are second-order. A 10% gain in LED system efficacy lowers cost by only ~1.5% for lettuce and ~3% for tomato at Current ($LUE_{INT} = 1.2$); comparable 10% gains in red and blue spectral efficacy yield ~1.4%/0.3% (lettuce) and ~2.7%/0.5% (tomato), respectively (see Section 3.1). As efficacies approach physical ceilings ([Kusuma et al., 2020](#)), diode improvements help but are unlikely to reduce production costs to benchmark levels on their own; greater leverage lies in biological improvements (higher LUE_{INT} , stronger sink capacity, cultivar development) and canopy architecture (LI).

Alongside operating energy, upfront investment cost is also mentioned as a structural constraint. Our CAPEX model is built bottom-up from KWIN ([Raaphorst and van Tuyll, 2023](#)) reference, adjusted to VF dimensions and complemented by VF-specific components across the categories Buildings, Lighting, Climate System, Water/Fertilisation/ CO_2 , Growing System, and Miscellaneous. This reliance on proven equipment and scale economies explains why the modelled capital cost per kilogram falls below the greenhouse benchmarks for most lettuce scenarios, and approaches the benchmark for tomato only under the highest performing design case.

A central limitation of this study is that the framework is designed to analyse biological and technical constraints and their implications for production cost, rather than to predict the detailed operation of a specific commercial vertical farm. Energy demand is calculated from representative photoperiod and dark-period balances for an average year and therefore does not resolve seasonal climate variation, changes in equipment efficiency at partial load, or short-term fluctuations in temperature and humidity control. In addition, the cost framework is parameterised for Dutch conditions to enable direct comparison with KWIN 2023 ([Raaphorst and van Tuyll, 2023](#)), including assumptions on electricity prices, labour costs, financing, and depreciation. Therefore, the absolute energy-use and cost estimates should be interpreted as indicative for the Dutch reference context, rather than as generally transferable cost predictions. The framework itself can be adapted to other crops, regions and price regimes by replacing crop parameters and local cost modules, but its quantitative outcomes should be re-evaluated under those specific conditions.

5 Conclusion

Cost prices for lettuce and tomato were modelled for a hypothetical Dutch vertical farm using a cost framework aligned with KWIN 2023 ([Raaphorst and van Tuyll, 2023](#)). Although parametrised for Dutch conditions, the framework itself is generic and can be adapted to other crops, production systems, and regions by replacing crop-specific PBM parameters and local cost modules. The physiological core remains unchanged: attainable yield is determined by the interaction between source and sink capacity, while absolute $\text{€ kg}^{-1} \text{ FW}$ outcomes depend on local energy prices, labour costs, financing conditions, and investment assumptions. The framework should therefore be interpreted as a tool for identifying crop-performance targets required for financially viable vertical farming rather than for predicting realised commercial outcomes. Future research should

validate and refine the framework across a wider range of crops, cultivation systems, and geographical contexts.

The results show which improvements in crop performance are required to reduce vertical farm production costs. Increasing source capacity through higher DLI can raise yield, but rapidly exposes sink limitation, making sink capacity and crop cycle time primary system constraints. For lettuce, combinations of improved physiological efficiency and crop performance were associated with modelled cost levels comparable to the Dutch greenhouse benchmark. For tomato, benchmark cost levels were not reached within the evaluated scenario space. The closest approaches to the greenhouse benchmark required both an accelerated development at elevated temperature and sustained high intercepted LUE—conditions that are not currently realised at full-cycle scale and therefore constitute major research and breeding challenges.

The future of VF depends on value for money. Further cost reduction will depend less on incremental hardware improvements and more on knowledge development and integration: understanding plant physiological limits, designing controllable environments that support those limits, and translating that understanding into reproducible cultivation recipes. Technology must therefore be developed explicitly in service of plant physiology. The scenarios evaluated here should be read as research targets—ambitious but not physically arbitrary—and provide quantitative guidance for experimental research, crop performance and cultivar development in controlled-environment agriculture.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding authors.

Author contributions

GM: Writing – original draft, Investigation, Resources, Formal analysis, Data curation, Project administration, Conceptualization, Validation, Methodology. DH: Writing – original draft, Investigation, Resources, Formal analysis, Data curation, Project administration, Conceptualization, Validation, Methodology. AS: Methodology, Investigation, Writing – original draft. JM: Conceptualization, Writing – review & editing, Supervision. JE: Writing – review & editing, Conceptualization, Supervision. PB: Supervision, Conceptualization, Writing – review & editing.

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Conflict of interest

GM is listed as an inventor on multiple patents related to vertical farming systems. He manages a company focused on the construction and operation of vertical farms: Seven Steps to Heaven, based in the Netherlands. DH is a research assistant at Seven Steps to Heaven. TS is an independent advisor to Seven Steps to Heaven. GM was employed by Seven Steps To Heaven B.V.

The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fsufs.2026.1833809/full#supplementary-material>

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